Homework 3
Out: March 14th, 2026, Saturday -- Due: April 1st, 2026, Wednesday, 11:59pm
EC327 Introduction to Software Engineering – Spring 2026

Total: 100 points

Submission

- For Q1 work with your TF do show them your fixed midterm during OH or Lab. They will sign you off. It will basically be "all or nothing" grading. You have unlimited attempts.
- For Q2, there will be a multiple-choice area of Gradescope for you to fill out.
- For Q3 write your answers clearly in your favorite text editor or word processor and submit a **pdf** file to Gradescope.
    - Username_HW2_Q3.pdf
- For Q4 make a picture however you like. Convert that to a PDF and upload it to Gradescope.
    - Username_HW2_Q4.pdf
- In summary
    - Q1 – work with TF
    - Q2 – Multiple choice on Gradescope
    - Q3 – PDF to Gradescope
    - Q4 – PDF to Gradescope

Failure to follow submission process will cost you points.

Q1. [25 pts]
Fix your midterm exam. Your TAs will get you back your submission. **We will post sample outputs/tests to Blackboard or Piazza**. Your code must (1) compile, (2) match this output, and (3) follow the spirit of the midterm (e.g., don't completely recode the whole thing to make it match the output but not do what the question originally asked).

Q2. [30 points; 1pts each]
Simply list each number and the letter for your solution.

1) In C++, new types are mostly created using _____.
a. a function
b. a class or a struct
c. either a or b
d. neither a nor b

2) In object-oriented programming, a data member plays the role of _____.
a. an attribute
b. a behavior
c. either a or b
d. neither a nor b

3) In object-oriented programming, a member function plays the role of a _____.
a. an attribute
b. a behavior
c. either a or b
d. neither an nor b

4) In C++, attributes and behaviors are declared in the _____.

a. class declaration
b. class definition
c. either a or b
d. neither an nor b

5) In C++, attributes and behaviors are normally defined in the _____.
a. class definition
b. member function definition
c. either a or b
d. neither an nor b

6) In a class, all members are by default_____.
a. private
b. protected
c. public
d. either a, b, or c

7) In C++, we need a _____ at the end of class definition.
a. colon
b. dot
c. semicolon
d. either a, b, or c

8) In a class, a private data member can be accessed _____.
a. in the class
b. through the public member functions
c. either a or b
d. neither a nor b

9) In a class, a public member function can be accessed _____.
a. in the class.
b. through other public member functions
c. outside the class
d. either a, b, or c

10) To define a member function outside of a class, we need the _____ operator.
a. member select (.)
b. class scope (::)
c. either a or b
d. neither a nor b

11) To call a member function outside of a class, we need the _____ operator.
a. member select (.)
b. class scope (::)
c. either a or b
d. neither a nor b

12) In a class, data members are normally initialized by _____.
a. a constructor
b. a destructor
c. either a or b
d. neither a nor b

13) In a class, the data members are normally cleared by _____.

a. a constructor
b. a destructor
c. either a or b
d. neither a nor b

14) A constructor is a member function that _____ as the class.

a. has the same name
b. cannot have a return value
c. neither a nor b
d. both a and b

15) A destructor is a member function that has _____ as the class.

a. the same name as the class preceded by a tilde
b. cannot have a return value
c. neither a nor b
d. both a and b

16) A class can have more than one _____.

a. default constructor
b. parameter constructor
c. destructor
d. assignment operator

17) In a class the _____ can be overloaded.

a. default constructor
b. parameter constructor
c. copy constructor
d. assignment operator

18) The initialization list in a constructor starts with a _____

a. dot
b. colon
c. semicolon
d. none of the above

19) If we create n objects in our application, _____ constructors are called.

a. 0
b. 1
c. n
d. none of the above

20) If we create n objects in our application, _____the destructor of the class needs to be called _____ times.

a. 0
b. 1
c. n
d. none of the above

21) We need _____ copy constructor
a. at least one
b. at most one
c. either a or b
d. both a and b

22) We can have _____ destructor (s) for the class.
a. only one
b. only two
c. any number of
d. no

23) If a class has three instance data members and we create five object of the class, the system creates _____ data values.
a. five
b. fifteen
c. one
d. no

24) If a class has two static data members and we create five object of the class, the system creates _____ static data values.
a. five
b. ten
c. two
d. no c

25) In a member function, a host object is _____.
a. the first parameter
b. the second parameter
c. a hidden parameter pointed by the this pointer
d. the return object

26) A mutator function is the one that _____ the host object.
a. changes
b. does not change
c. a or b
d. neither a nor b

27) An accessor function is the one that _____ the host object.
a. changes
b. does not change
c. a or b
d. neither a nor b

28) One or more conditions that may need to be imposed on some or all class data members is the _____ of the class.
a. class definition
b. class conditions
c. class invariants
d. class declaration

29) A static data member needs to be initialized _____.

a. in the class declaration
b. in the class definition
c. in the global area
d. none of the above

30) A static member function can be called through _____.

a. an instance
b. the class
c. either a or b
d. neither a nor b

Q3. Explain the following compiler errors and why they might happen [24 pts; 2pt each]

1. "Stack overflow"
2. "Undefined reference to main" Hint: many of you see this when you break code up into various .cpp files.
3. "`fooVar' undeclared (first use this function)"
4. foo.h: No such file or directory
5. "unterminated string or character constant"
6. "ANSI C++ forbids comparison between pointer and integer"
7. "Null pointer exception" or "Segmentation Fault"
8. `cout' undeclared (first use this function)
9. parse error before `something'
10. "'member' is not accessible"
11. "Destructor name must match the class name"
12. "Friends must be functions or classes"

Q4. UML Class Diagram [21 pts]

Create the UML class diagram for the classes in **PA3**. Minimally indicate the class inheritance hierarchy, public, private, and protected member functions/variables, and class name. Your diagram should be organized and easy to follow. Please consider using UML creation software in-lieu of a hand drawn diagram (ArgoUML for example). References:

http://en.wikipedia.org/wiki/Class_diagram
https://argouml-tigris-org.github.io/tigris/argouml/